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Semester: III

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Subject: Programming with Java

ASSIGNMENT No : 1

TITLE : Java Environment Setup and Basic Program Execution

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyse outputs, and apply the concepts to solve practical programming problems.

Programme

Output

```
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA> cd basic
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\basic> javac Calculator.java
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\basic> java Calculator
Addition: 49
Subtraction: 31
Multiplication: 360
Division: 4
PS C:\Users\dhruv\OneDrive\Pictures\Documents\Desktop\Study\sem 3\JAVA\basic> |
```

Conclusion

Questions

1. Explain the main concept used in Java environment setup, variables and methods.

Ans-- The Java environment setup is built around three layered components: the JDK (development tools), the JRE (runtime libraries), and the JVM (execution engine). Together they allow a developer to write, compile, and run Java code, with the JVM ensuring platform independence by executing bytecode on any operating system.

2. What are the advantages of using Java variables and methods?

Ans-- Java variables provide a way to store and manage data efficiently, while methods promote code reusability, modularity, and readability. They simplify program maintenance, reduce code duplication, and improve the overall organization and performance of Java applications.

3. How can this implementation be improved?

Ans-- The implementation can be improved by following object-oriented principles such as encapsulation, inheritance, and polymorphism. Using meaningful variable and method names, proper exception handling, comments, and code optimization also enhances readability, maintainability, and efficiency.

4. What are the real-world applications?

Ans--JDK/JRE/JVM architecture powers cross-platform enterprise software, Android app development (via similar runtime concepts), and large-scale distributed systems like banking and e-commerce platforms. Its portability and memory management make it a preferred choice for building scalable, long-running server applications.